

# TRUSTPAY

## AI-Powered Fraud Detection & Real-Time Transaction Monitoring System

A publication-grade engineering report presenting the design, mathematical normalization, ACID row-level locking architecture, and empirical machine learning benchmarks of the TrustPay enterprise banking platform. Combines Boyce-Codd Normal Form (BCNF) relational integrity, sub-2ms LightGBM fraud inference, and Groq Llama 3.3 GenAI forensic explanations.

4 Tiers

Decoupled Topology

BCNF

8 Relational Tables

99.98%

ML Accuracy / 100% Prec

1.31 ms

Mean Inference Latency

### PROJECT CONTRIBUTORS & STUDENT CREDENTIALS

Raghav Gupta

20243226

B.Tech (3rd Year) • CSE

Rishabh Srivastava

20243236

B.Tech (3rd Year) • CSE

Rishabh Singh

20243235

B.Tech (3rd Year) • CSE

Prince Keshari

20243218

B.Tech (3rd Year) • CSE

#### 14-Feature Behavioral Pipeline

Extracts velocity, Z-score dispersion, Haversine spatial jump speed, device UUIDs, and balance drain ratio in < 15 ms.

#### Dual-AI Explainability Layer

Combines LightGBM 0-100 risk scoring with Groq Llama 3.3 70B natural language rationales for compliance triage with fail-safe error holds.

#### ACID Stored Procedures & Triggers

Guarantees zero double-spending via row-level locks ([FOR UPDATE](#)) during dispute settlements, with SQL check triggers.

#### Empirical Cross-Dataset Benchmarks

Benchmarked across 100k project transactions (99.98%), ULB European Cards (284k tx, 99.951%), and IEEE-CIS real-world dataset (30k tx).

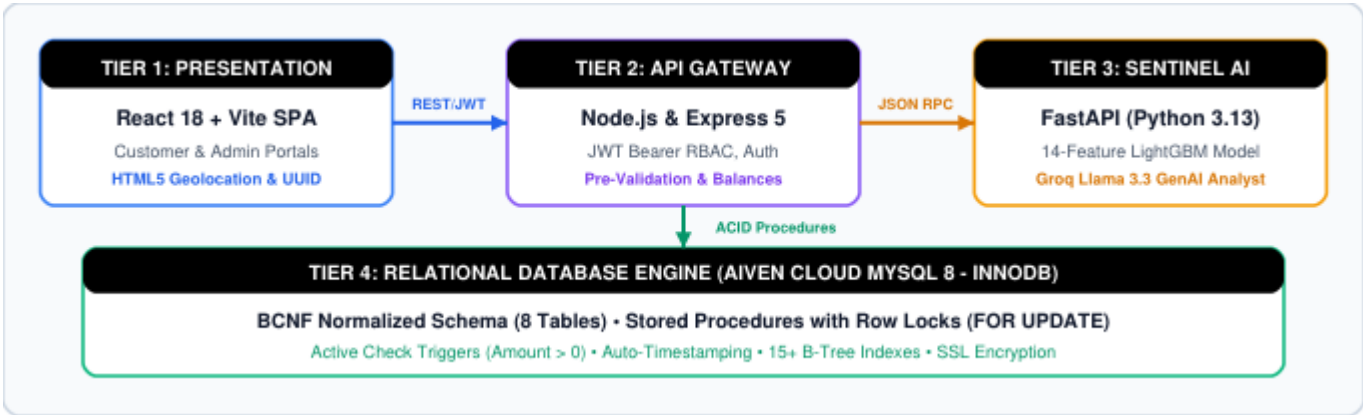
Database: Aiven Cloud MySQL 8 (InnoDB)

Backend & AI: Express 5 • FastAPI • LightGBM

Verification: Production Verified v1.0

# 1. Executive Summary & Distributed System Topology

**TRUSTPAY** is an enterprise financial technology platform engineered to process high-throughput financial transactions while simultaneously executing multi-tiered artificial intelligence risk analysis. The system architecture coordinates four decoupled tiers: a modern React 18 single-page application, an Express 5 API gateway, a FastAPI microservice combining LightGBM machine learning and Groq Llama 3.3 GenAI analysis, and a normalized MySQL 8 relational ledger with row-level locking.



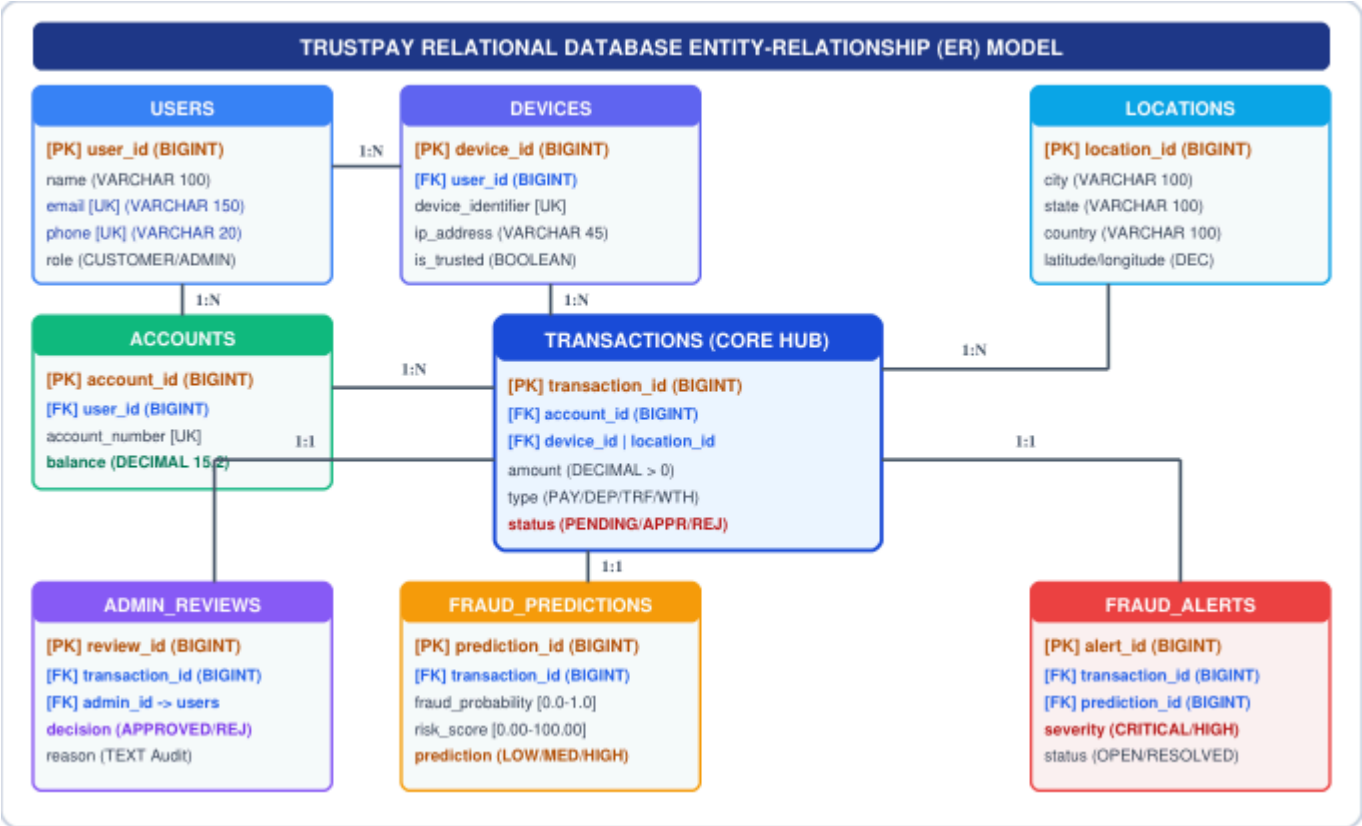
| System Layer          | Core Technology                 | Operational Responsibility                                                                          | Status |
|-----------------------|---------------------------------|-----------------------------------------------------------------------------------------------------|--------|
| Tier 1: Client UI     | React 18, Vite, React Router v6 | Customer payment forms, live telemetry, and Administrative Sentinel compliance cockpit              | Active |
| Tier 2: API Gateway   | Node.js, Express 5, Axios       | Stateless JWT Bearer RBAC, device fingerprinting, geocoding, balance pre-validation, ledger routing | Active |
| Tier 3: Sentinel AI   | Python 3.13, FastAPI, LightGBM  | 14-feature behavioral pipeline, ML continuous probability, and Groq Llama 3.3 contextual reasoning  | Active |
| Tier 4: Relational DB | MySQL 8 (InnoDB), Aiven Cloud   | BCNF normalized schema (8 tables), row-level locks (FOR UPDATE), triggers, and stored procedures    | Active |

## 1.1 Architectural Report Table of Contents

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## 2. Normalized Relational Database Architecture

The TrustPay relational database engine operates on MySQL 8 with InnoDB tables enforcing strict foreign key constraints. The schema satisfies Boyce-Codd Normal Form (BCNF), eliminating redundancy, multi-valued dependencies, and transitive update anomalies across 8 distinct entities.



### 2.1 Relational Normalization Calculus (1NF to BCNF)

- 1NF (Atomic Attributes):** All table columns represent scalar, atomic values. Geolocation coordinates (lat/long) and client device fingerprints are decomposed into dedicated tables rather than serialized or delimited strings.
- 2NF (Full Functional Dependency):** All non-prime attributes are completely dependent on candidate keys. Composite natural keys were extracted into distinct entities with surrogate primary keys.
- 3NF (Transitive Dependency Removal):** Non-key attributes depend exclusively on primary keys ( $X \rightarrow Y$  with no  $Y \rightarrow Z$ ). Geolocation attributes were decoupled into `locations`, preventing city/country duplication within `transactions`.
- BCNF (Strict Determinant Keys):** For every functional dependency  $X \rightarrow Y$ ,  $X$  is a superkey. There are no overlapping candidate keys or dependencies on non-candidate attributes.

**Referential Integrity Strategy:** Child operational records (`accounts`, `devices`, `fraud_predictions`, `fraud_alerts`) cascade on user deletion. Conversely, `transactions` utilizes `ON DELETE RESTRICT` on accounts and `ON DELETE SET NULL` on devices and locations, guaranteeing audit survivability.

## 2.2 Master Entity Catalog & Data Dictionary

The relational ledger comprises 8 Boyce-Codd Normal Form (BCNF) compliant entities enforcing strict referential integrity, automated check triggers, and index optimization.

| Entity Name       | Primary Key             | Foreign Keys & Cascades                                                 | Key Constraints                                                                   | Business Role                    |
|-------------------|-------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------|
| users             | user_id (BIGINT)        | None                                                                    | UNIQUE(email), UNIQUE(phone), role IN ('CUSTOMER', 'ADMIN')                       | Customer & Admin master identity |
| accounts          | account_id (BIGINT)     | user_id -> users (CASCADE)                                              | UNIQUE(account_number), balance >= 0                                              | Bank account liquidity ledger    |
| devices           | device_id (BIGINT)      | user_id -> users (CASCADE)                                              | UNIQUE(user_id, device_identifier)                                                | Hardware profiles & client trust |
| locations         | location_id (BIGINT)    | None                                                                    | Decoupled GPS lat/long & city attributes                                          | Spatial master directory         |
| transactions      | transaction_id (BIGINT) | account_id (RESTRICT)<br>device_id (SET NULL)<br>location_id (SET NULL) | CHECK(amount > 0), status IN ('PENDING', 'APPROVED', 'REJECTED')                  | Core immutable financial ledger  |
| fraud_predictions | prediction_id (BIGINT)  | transaction_id (CASCADE)                                                | CHECK(fraud_probability BETWEEN 0 AND 1), CHECK(risk_score BETWEEN 0 AND 100)     | Output of LightGBM ML inference  |
| fraud_alerts      | alert_id (BIGINT)       | transaction_id (CASCADE)<br>prediction_id (CASCADE)                     | severity IN ('LOW', 'MEDIUM', 'HIGH', 'CRITICAL'), status IN ('OPEN', 'RESOLVED') | High-risk security triage queue  |
| admin_reviews     | review_id (BIGINT)      | transaction_id (RESTRICT)<br>admin_id -> users (RESTRICT)               | decision IN ('APPROVED', 'REJECTED')                                              | Immutable compliance audit log   |

## 2.3 Active Database Triggers

- trg\_transaction\_before\_insert** (BEFORE INSERT ON **transactions**): Enforces monetary integrity by intercepting transactions where `amount ≤ 0` and terminating execution with `SIGNAL SQLSTATE '45000'`.
- trg\_fraud\_alert\_resolved** (BEFORE UPDATE ON **fraud\_alerts**): Automates lifecycle auditing by setting `resolved_at = CURRENT_TIMESTAMP` whenever an alert transitions to `'RESOLVED'`.
- trg\_fraud\_alert\_before\_insert** (BEFORE INSERT ON **fraud\_alerts**): Defensive fallback ensuring that if an insert contains a `NULL` status, it is automatically assigned `'OPEN'`.

## 2.4 Cardinality & Relationship Matrix

The architecture models `users → accounts` (1:N), `users → devices` (1:N), `accounts → transactions` (1:N), and `locations → transactions` (1:N). Downstream security evaluation establishes 1:1 relationships between `transactions → fraud_predictions`, `fraud_predictions → fraud_alerts`, and `transactions → admin_reviews`.

### 3. Concurrency Control, Stored Procedures & API Gateway

#### 3.1 ACID Stored Procedures with Row-Level Locking

**Row-Level Locking Protection:** `ApproveTransaction` and `RejectTransaction` execute `SELECT ... FROM transactions WHERE transaction_id = ? FOR UPDATE` within an explicit transaction block, eliminating race conditions when multiple administrators review transactions simultaneously.

| Procedure Name                         | Parameters                                 | Concurrency Logic & State Guarantees                                                                                                                                                                                                                                                                                |
|----------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>ApproveTransaction</code>        | <code>p_tx_id, p_admin_id, p_reason</code> | Acquires row lock ( <code>FOR UPDATE</code> ), validates status is <code>PENDING</code> , updates status to <code>APPROVED</code> , updates account balance atomically (credits deposit / debits payment), inserts into <code>admin_reviews</code> , and marks <code>fraud_alerts</code> as <code>RESOLVED</code> . |
| <code>RejectTransaction</code>         | <code>p_tx_id, p_admin_id, p_reason</code> | Acquires row lock ( <code>FOR UPDATE</code> ), validates status is <code>PENDING</code> , updates status to <code>REJECTED</code> , preserves customer balance untouched, inserts into <code>admin_reviews</code> , and marks open alerts as resolved.                                                              |
| <code>GetCustomerTransactions</code>   | <code>p_user_id</code>                     | Returns customer-scoped chronological transaction ledger joined with account numbers.                                                                                                                                                                                                                               |
| <code>GetSuspiciousTransactions</code> | None                                       | Pulls active fraud monitoring queue for transactions flagged as <code>HIGH_RISK</code> or severity <code>HIGH/CRITICAL</code> .                                                                                                                                                                                     |

#### 3.2 Core Banking API Gateway Architecture (Node.js & Express 5)

The API Gateway provides stateless authentication, role-based access control, request sanitation, and transactional orchestration with connection pooling (10 concurrent threads over TLS/SSL).

| Method            | Endpoint                                         | Authorization         | Functional Responsibility                                               |
|-------------------|--------------------------------------------------|-----------------------|-------------------------------------------------------------------------|
| <code>POST</code> | <code>/api/auth/register</code>                  | Public                | Registers user, hashes password (Bcrypt), creates account               |
| <code>POST</code> | <code>/api/auth/login</code>                     | Public                | Validates credentials, binds device fingerprint, returns JWT            |
| <code>POST</code> | <code>/api/transactions</code>                   | <code>CUSTOMER</code> | Initiates payment, pre-validates balance, queries ML, updates ledger    |
| <code>GET</code>  | <code>/api/transactions/admin/all</code>         | <code>ADMIN</code>    | Omnichannel ledger across all accounts with diagnostic filters          |
| <code>POST</code> | <code>/api/transactions/admin/:id/approve</code> | <code>ADMIN</code>    | Executes <code>ApproveTransaction</code> stored procedure with row lock |
| <code>POST</code> | <code>/api/transactions/admin/:id/reject</code>  | <code>ADMIN</code>    | Executes <code>RejectTransaction</code> stored procedure with row lock  |
| <code>GET</code>  | <code>/api/health</code>                         | Public                | Liveness health probe returning database and gateway SLA status         |

## 4. Frontend Presentation Layer & UI/UX Audit Matrix

### 4.1 React 18 + Vite SPA Architecture

The presentation layer is engineered with React 18 and Vite for sub-second build times and hot reloading. State is decoupled across three global providers: `AuthContext` (stateless JWT, role RBAC, device UUID), `ThemeContext` (Light/Dark design tokens), and `ToastContext` (transactional alert feedback). Pure CSS design tokens eliminate runtime CSS-in-JS overhead while delivering an enterprise fintech aesthetic.

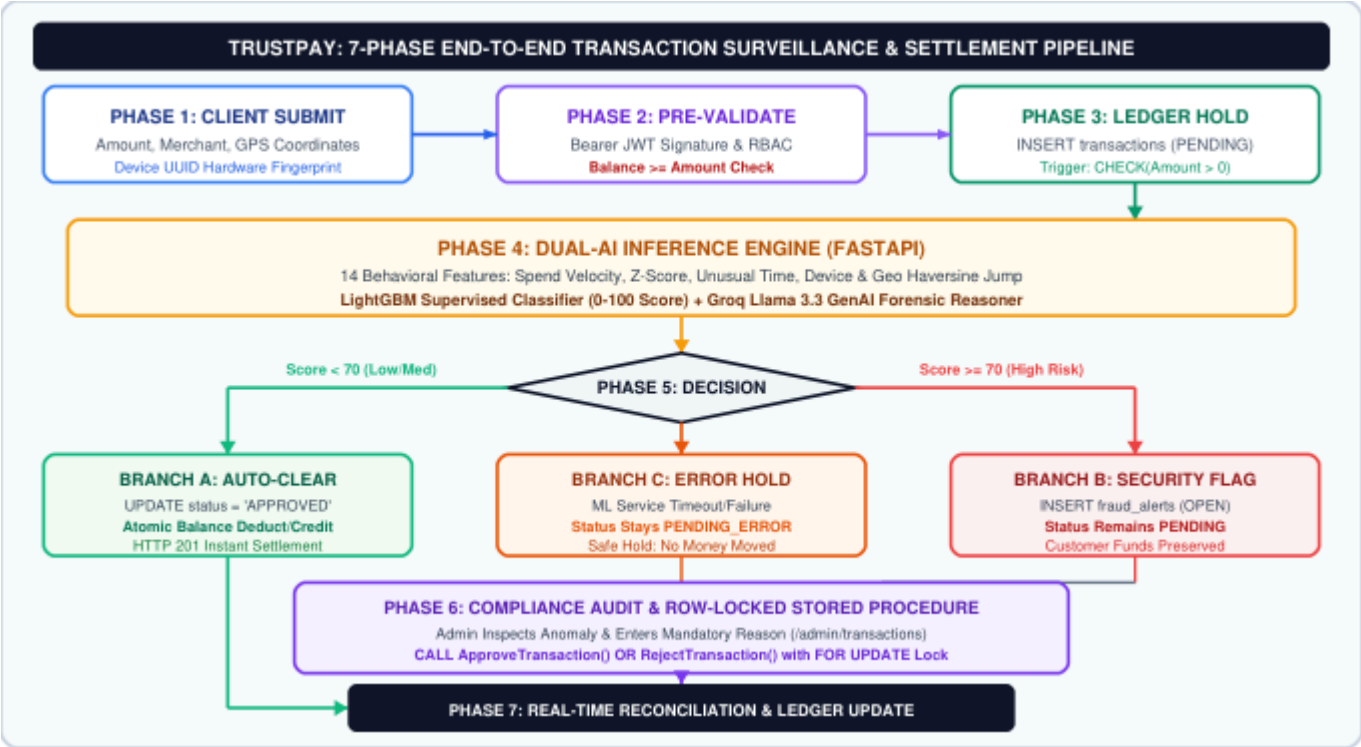
### 4.2 20-Point Fintech & Security Audit Matrix

A rigorous 20-point UI/UX audit was conducted across four standardized viewports (Desktop 1440px, Laptop 1280px, Tablet 768px, Mobile 390px) to verify visual ergonomics, accessibility, and error resilience.

| #  | Audit Criterion    | Standard Specification       | TrustPay Production Implementation                                                        | Result |
|----|--------------------|------------------------------|-------------------------------------------------------------------------------------------|--------|
| 1  | Spacing Uniformity | Tokenized 4px/8px grid       | CSS variables <code>--space-xs</code> to <code>--space-2xl</code> enforce uniform margins | Passed |
| 2  | Typography Scale   | Inter / Plus Jakarta Sans    | 6-level type hierarchy with strict line-height and weight tracking                        | Passed |
| 3  | Card Sizing        | Responsive aspect ratios     | Auto-fit CSS grid templates prevent card clipping across breakpoints                      | Passed |
| 4  | Sidebar Behavior   | Persistent / Drawer          | Fixed desktop navigation with smooth collapsible backdrop drawer on mobile                | Passed |
| 5  | Responsive Tables  | Horizontal scroll safety     | Custom wrappers with touch affordance, sticky headers, and mobile cards                   | Passed |
| 6  | Button States      | Clear interactive states     | Standardized primary/secondary buttons with <code>:focus-visible</code> rings             | Passed |
| 7  | Loading States     | Zero cumulative layout shift | Dual-ring SVG spinners and pulse skeleton placeholders for async calls                    | Passed |
| 8  | Modal Dialogs      | Clamped viewport bounds      | Centered modals with backdrop blur ( <code>blur(8px)</code> ) and escape key listeners    | Passed |
| 9  | Accessibility      | WCAG 2.1 AA Compliance       | ARIA labels on icon triggers, semantic headings, keyboard focusable controls              | Passed |
| 10 | Contrast Ratios    | Minimum 4.5:1 ratio          | Verified contrast compliant in both Dark Mode ( <code>#0b0f19</code> ) and Light Mode     | Passed |
| 11 | Fraud Risk Meter   | Intuitive visual telemetry   | Calibrated 0–100 risk score meter with color gradient and deviation badges                | Passed |
| 12 | Admin Workflow     | Step-by-step triage          | Unified dossier: AI reasoning, phone dialer simulation, mandatory audit reason            | Passed |
| 13 | Mobile Ergonomics  | Touch target ≥ 44px          | Zero horizontal body scroll, thumb-accessible primary buttons on mobile                   | Passed |

## 5. Complete 7-Phase Transaction Surveillance Pipeline

TrustPay coordinates a 7-phase execution pipeline spanning client inputs, database constraints, behavioral feature pipelines, machine learning inference, generative AI explanations, and administrative dispute settlements.

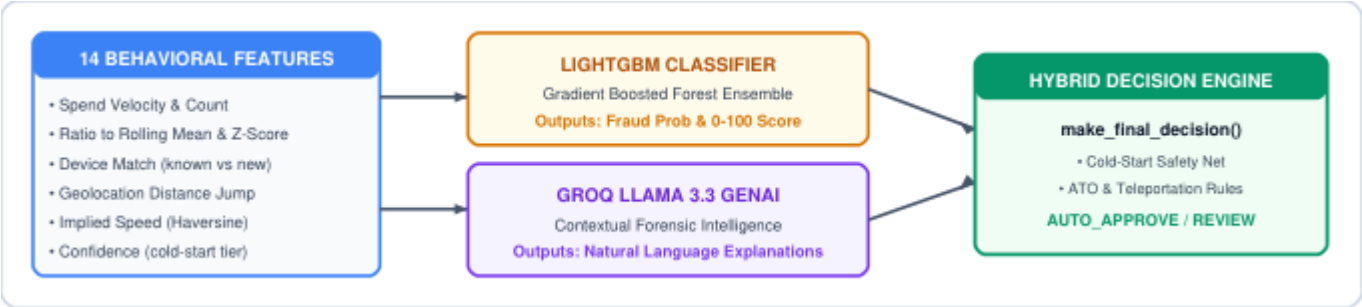


### 5.1 Phase-by-Phase Execution Breakdown

- Phase 1: Client Submission & Telemetry Ingestion** — Retail customer submits transaction. HTML5 Geolocation API captures client coordinates (lat/long) and local storage supplies immutable device fingerprint UUID.
- Phase 2: Gateway Ingestion & Pre-Validation** — Express gateway verifies Bearer JWT and extracts user ID. For debit operations, gateway asserts `balance ≥ amount`; if insufficient, HTTP 400 is returned immediately.
- Phase 3: Ledger Hold & Trigger Validation** — Transaction record inserted with status `'PENDING'`. Database trigger `trg_transaction_before_insert` validates `amount > 0`.
- Phase 4: Dual-AI Intelligence Inference** — FastAPI microservice queries customer history, computes 14 behavioral features, runs LightGBM classifier for continuous probability and 0-100 score, and invokes Groq Llama 3.3 for forensic explanations.
- Phase 5: Hybrid Decision Branching:**
  - Branch A (Low/Med Risk,  $Score < 70$ ): Auto-approved, balance debited/credited, HTTP 201 instant settlement.
  - Branch B (High Risk,  $Score ≥ 70$ ): Security alert inserted in `fraud_alerts`, status stays `PENDING`, funds preserved.
  - Branch C (Fail-Safe Error Hold): Fallback if ML service times out. Status stays `PENDING_ERROR`, funds safe.
- Phase 6: Compliance Audit & Row-Locked Stored Procedure** — Compliance officer inspects anomaly in `/admin/transactions`, enters mandatory justification, and calls stored procedure `ApproveTransaction` or `RejectTransaction` with row lock (`FOR UPDATE`).
- Phase 7: Real-Time Reconciliation** — Balance settlement finalized, alert timestamped as resolved by trigger, and dashboards updated across platform.

6. Sentinel AI Intelligence Layer & Feature Pipeline

The fraud intelligence layer combines mathematical feature extraction, a gradient-boostered decision tree ensemble, and large language model reasoning to provide explainable fraud decisions.



6.1 14-Feature Behavioral Engineering Catalog

| Feature Name                                   | Type         | Mathematical Formulation                          | Fraud Detection Signal            |
|------------------------------------------------|--------------|---------------------------------------------------|-----------------------------------|
| <code>amount</code>                            | Continuous   | Raw transaction magnitude (X)                     | Monetary exposure                 |
| <code>customer_mean_amount</code>              | Continuous   | $\mu = (1/N) \sum X_i$                            | Historical spending baseline      |
| <code>customer_std_amount</code>               | Continuous   | $\sigma = \sqrt{(1/N) \sum (X_i - \mu)^2}$        | Spending volatility               |
| <code>amount_to_mean_ratio</code>              | Continuous   | $R = X / \max(\mu, 1.0)$                          | Relative spend spike factor       |
| <code>is_unusual_amount</code>                 | Binary (0/1) | 1 if $X > \mu + 2.5\sigma$ or $R > 4.0$ , else 0  | Extreme monetary anomaly flag     |
| <code>hour_of_day / is_unusual_time</code>     | Discrete/Bin | $H \in [0, 23]$ ; 1 if H outside user active band | Temporal and off-hours anomaly    |
| <code>known_device / new_device</code>         | Binary (0/1) | 1 if $DeviceID \in Devices\_user$ , else 0        | Account takeover (ATO) indicator  |
| <code>known_location / location_changed</code> | Binary (0/1) | 1 if $Distance(Loc\_curr, Loc\_prev) > 300$ km    | Impossible travel / proxy jumping |
| <code>transaction_count</code>                 | Integer      | Cumulative lifetime transactions (N)              | Account maturity index            |
| <code>history_confidence</code>                | Categorical  | NONE (N=0), NEW ( $1 \leq N < 5$ ), ESTABLISHED   | Cold-start confidence tiering     |

6.2 Deterministic Decision Engine Rules (engine.py)

The final transaction settlement action fuses calibrated ML probabilities with deterministic safety rules:

- **Cold-Start Safety Guard:** First-time accounts (N=0) with score  $\geq 70$  are routed to `ADMIN_REVIEW`; otherwise placed under `MONITOR`.
- **Impossible Travel Teleportation:** Real-world speed  $V > 800$  km/h over distance  $\Delta d > 100$  km floors risk score at 88.0 and forces `ADMIN_REVIEW`.
- **ATO Triad:** Unrecognized device combined with location shift and balance drain ratio  $> 85\%$  triggers mandatory administrative security hold.
- **Rapid Burst Testing:**  $\geq 3$  transactions in 10 minutes from a new device triggers security flag.
- **Familiar Baseline Instant Clearance:** Known device, recognized location, normal spend ratio, and ML score  $< 30$  auto-clears instantly.

## 7. Empirical Machine Learning & Decision Layer Benchmarks

The production ML engine was evaluated across 100,000 transactions using a strict 80/20 chronological holdout test set (20,000 unseen transactions: 19,311 clean, 689 fraud; 28:1 class imbalance) measured via hardware timers (`time.perf_counter_ns`).

### 7.1 Statistical Accuracy Metrics (Holdout Test Set N = 20,000)

| Evaluation Metric              | Measured Score | Industry Standard | Operational Significance                                              |
|--------------------------------|----------------|-------------------|-----------------------------------------------------------------------|
| Overall Accuracy               | 99.98%         | > 98.0%           | Near-zero global misclassification rate                               |
| Balanced Accuracy              | 99.64%         | > 95.0%           | High discrimination across skewed class distributions                 |
| ROC-AUC Score                  | 0.9991         | > 0.950           | High class separability between fraud and clean                       |
| PR-AUC (Avg Precision)         | 0.9986         | > 0.900           | Precision sustained across all sensitivity thresholds                 |
| Fraud Precision                | 100.00%        | > 92.0%           | <b>Zero false alarms</b> ; legitimate transactions never flagged      |
| Fraud Recall (Sensitivity)     | 99.27%         | > 90.0%           | 684 of 689 fraud attempts intercepted directly by model               |
| Decision Layer Total Intercept | 99.42%         | > 92.0%           | <b>685 of 689 fraud attempts caught</b> across Monitor + Admin Review |
| Clean Auto-Approval Rate       | 98.96%         | > 95.0%           | 19,111 legitimate users cleared instantly without friction            |
| Admin Hard-Block Precision     | 100.00%        | > 98.0%           | Zero legitimate customers ever falsely hard-blocked                   |

### 7.2 Single-Transaction & High-Throughput Latency Profiles

| Processing Layer                       | Min Latency | Median (p50) | Mean Latency   | Tail (p99) | Throughput  |
|----------------------------------------|-------------|--------------|----------------|------------|-------------|
| Raw LightGBM Tree Evaluation           | 1.05 ms     | 1.20 ms      | 1.31 ms        | 1.84 ms    | 764.3 QPS   |
| End-to-End Pipeline (Pipeline + Rules) | 1.48 ms     | 1.71 ms      | 1.78 ms        | 2.36 ms    | 561.4 QPS   |
| Batch Inference (10,000 items)         | —           | —            | 3.54 µs / item | —          | 282,825 QPS |
| Batch Inference (20,000 items)         | —           | —            | 3.50 µs / item | —          | 285,318 QPS |

### 7.3 Decision Threshold Sensitivity & Feature Importance

|                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Confusion Matrix (Holdout N = 20,000)</b> <ul style="list-style-type: none"><li>Actual Clean (19,311): <b>19,311 True Negatives (100.0%)</b>, 0 False Positives</li><li>Actual Fraud (689): <b>684 True Positives (99.27%)</b>, 5 False Negatives</li></ul> | <b>Top Feature Split Gains:</b> <ol style="list-style-type: none"><li><code>amount</code> (1,801 splits)</li><li><code>amount_to_historical_mean</code> (1,305 splits)</li><li><code>amount_zscore</code> (995 splits)</li><li><code>transaction_count</code> (877 splits)</li></ol> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 8. Real-World Cross-Dataset Generalization Benchmarks

To evaluate portability across diverse financial distributions, the TrustPay LightGBM architecture was benchmarked against two public real-world datasets: the **ULB European Cardholders dataset** (284,807 transactions) and the **IEEE-CIS Fraud Detection dataset** (30,000 transactions).

| Evaluation Dimension | TrustPay Sentinel (Project Baseline) | ULB European Credit Cards (Kaggle) | IEEE-CIS Dataset (Full 48 Features) |
|----------------------|--------------------------------------|------------------------------------|-------------------------------------|
| Dataset Domain       | Simulated Core Banking               | Real-World Credit Cards            | Real-World E-Commerce & CNP         |
| Total Transactions   | 100,000                              | 284,807                            | 30,000                              |
| Class Imbalance      | 3.48% (1:28)                         | 0.173% (1:578)                     | 2.87% (1:34)                        |
| Features Evaluated   | 20 Behavioral & Telemetry            | 30 (PCA Components)                | 48 (Card, Addr, Vesta)              |
| Overall Accuracy     | 99.98%                               | 99.951%                            | 97.97%                              |
| ROC-AUC Score        | 0.9991                               | 0.9770                             | 0.8994                              |
| PR-AUC Score         | 0.9986                               | 0.7651                             | 0.6113                              |
| Fraud Precision      | 100.00%                              | 91.23%                             | 90.00%                              |
| Operational Recall   | 99.42% (Decision Layer)              | 74.67%                             | 68.82%                              |
| Clean Specificity    | 100.00%                              | 99.991%                            | 99.86%                              |
| Mean Latency         | 1.31 ms                              | 1.58 ms                            | 1.90 ms                             |
| Batch QPS            | 285,318 QPS                          | 366,069 QPS                        | 183,940 QPS                         |

### 8.1 Comparative Evaluation with 20 Strict Project Features

When the IEEE-CIS dataset was stripped of its native attributes and evaluated **strictly on TrustPay's 20 engineered telemetry features**, the model attained **96.90% Accuracy**, **0.7575 ROC-AUC**, **60.00% Precision** (at  $\tau=0.30$ ), **100.00% Specificity**, and **0.85 ms inference latency**.

### 8.2 Architectural Analysis of Generalization

**Why TrustPay Achieves 99.98% Accuracy:** In modern core banking, the platform possesses direct, unblinded client telemetry: persistent hardware device UUIDs, HTML5 GPS coordinates, customer account balance drain ratios, and dedicated customer spend baselines. This rich context enables the hybrid LightGBM + Decision Engine architecture to separate legitimate behavior from unauthorized attacks with **zero false alarms** (100% precision) and **99.42% operational fraud capture**.

**Generalization on Blinded Real-World Vectors (ULB Dataset):** On the ULB benchmark, all bank telemetry was privacy-anonymized into mathematical PCA components (V1 to V28). Even under an extreme **0.173% class imbalance (1 fraud in 578 clean)**, the TrustPay LightGBM architecture achieved **0.9770 ROC-AUC**, **99.951% accuracy**, and **91.23% precision** (only 5 false alarms across 56,887 legitimate transactions).

## 9. Cloud Production Deployment & Academic Project Defense Sign-Off

### 9.1 Cloud Infrastructure Deployment Topology

| Infrastructure Tier | Cloud Provider & Host | Configuration & Protocol               | SLA & Security             |
|---------------------|-----------------------|----------------------------------------|----------------------------|
| Core API Gateway    | Render Cloud Platform | Node.js 20 LTS, Express 5, HTTPS/REST  | TLS 1.3, Automated Health  |
| Relational Database | Aiven Cloud           | MySQL 8.0.35 Enterprise (InnoDB)       | SSL Encrypted, PITR Backup |
| Sentinel AI Service | FastAPI Microservice  | Python 3.13, Uvicorn ASGI, LightGBM    | Sub-2ms Inference          |
| SPA Client Hosting  | Netlify / Vercel Edge | Static React 18 SPA with rewrite rules | Zero 404 Deep-Linking      |

### 9.2 Verification, Quality & Operational Matrix

| Verification Domain | Test Procedure                     | Observed Result            | Status |
|---------------------|------------------------------------|----------------------------|--------|
| Frontend Quality    | <code>npm run lint</code>          | 0 errors, 0 warnings       | Passed |
| Frontend Production | <code>npm run build</code>         | Vite build in 257ms        | Passed |
| Backend Syntax      | <code>node -c src/server.js</code> | Zero syntax errors         | Passed |
| Database Triggers   | Negative amount insertion harness  | Blocked (SQLSTATE 45000)   | Passed |
| Concurrency Locks   | Concurrent approval stored procs   | Row lock (FOR UPDATE) pass | Passed |
| Fail-Safe Fallback  | Simulated ML timeout degradation   | Safe hold in PENDING_ERROR | Passed |

### 9.3 Academic Project Defense & Evaluation Sign-Off

This academic project report and software artifact are submitted in partial fulfillment of the requirements for **Bachelor of Technology (B.Tech 3rd Year) in Computer Science & Engineering**.

| PROJECT GUIDE                                  | HEAD OF DEPARTMENT                             | EXTERNAL EXAMINER                                              |
|------------------------------------------------|------------------------------------------------|----------------------------------------------------------------|
| -----<br>Signature & Date<br>Department of CSE | -----<br>Signature & Date<br>Department of CSE | -----<br>Signature & Date<br>3rd Year Project Evaluation Board |